Sankarshana Venugopal

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Education

Yonsei University Sep 2023 – Feb 2026
M.S. in Artificial Intelligence | Advised by Professor Jonghyun Choi *Seoul, South Korea*

Yonsei University Mar 2019 – Aug 2023
B.S. in Computer Science *Seoul, South Korea*

Publications

Sankarshana V, Mohammad Mostafavi, Jonghyun Choi
Under Review: A Novel Continuous-time Bridge Flow-Map Model Theory for Image-to-Image Translation
Submitted at International Conference on Learning Representations, 2027

- Architected and submitted a novel generative model theory that generalizes continuous-time Consistency Bridge Models and Diffusion Bridge Flow-Map Models, achieving state-of-the-art image-to-image translation performances in just 2 steps.
- Paper currently undergoing double-blind peer review.

Sankarshana V, Mohammad Mostafavi, Jonghyun Choi
DBMSolver: A Training-free Diffusion Bridge Sampler for High-Quality Image-to-Image Translation
Accepted at Computer Vision and Pattern Recognition (CVPR) Main Track, 2026

- Mathematically derived and implemented a novel, training-free diffusion bridge sampler, converting complex mathematical derivations into an efficient sampling algorithm.
- Developed custom solvers to accelerate inference times by 5× to 20× for state-of-the-art image-to-image translation without requiring computationally expensive model retraining.

Sankarshana V, Mohammad Mostafavi, Jonghyun Choi
FALCON: Fast Adaptive Lightweight Computation of Intensities and Events for Depth Estimation
Accepted at Computer Vision and Pattern Recognition (CVPR) Findings Track, 2026

- Developed a novel stereo matching framework with Spiking Neurons specifically optimized for the high temporal resolution and sparsity of asynchronous event-based vision data.
- Architected robust data processing and feature extraction pipelines in PyTorch to accurately estimate depth and spatial geometries from event camera inputs.

Experience

Graduate Researcher Sep 2023 – Mar 2026

- Machine Perception and Reasoning Lab, Seoul National University.
- Advised by Prof. Jonghyun Choi.
- Spearheaded research in generative models and event-based vision.

Teaching Assistantships Mar 2021 – Feb 2026

- CCO2105 Introduction to AI Research.
  - For Prof. Youngwoon Lee, Fall 2024, Yonsei University.
- ECE430 Introduction to Random Variables and Processes.
  - For Prof. Jonghyun Choi, Spring 2024, Seoul National University.
- AAI3201 Theory and Practice of Deep Learning.

- For Prof. Jonghyun Choi, Fall 2023, Yonsei University.

- ENG1108 Engineering Information Processing.
  - For Prof. Yong Seob Lee, Fall 2023, Yonsei University.
  - Conducted 2-hour lectures teaching Python to freshmen every week, completely in Korean.

Undergraduate Research Internships Nov 2022 – Aug 2023

- Vision and Learning Lab, Yonsei University
- Advised by Prof. Jonghyun Choi.

Front-end Developer Internship Jun 2022 – Sep 2022

In charge of website development at Bobidi Inc. using JavaScript and NodeJS.

App Developer Internship Jun 2021 – Aug 2021

Developed and published Corp Via's cross-platform app using Flutter and NoSQL database.

Awards and Honors

Outstanding International Student Scholarship Sep 2023 – Feb 2026

Awarded by Yonsei University.

Global Korea Scholarship for Undergraduate Studies Mar 2018 – Aug 2023

Awarded by the Government of South Korea.

Best Graduation Thesis Project Award Jun 2022

Awarded by the Department of Artificial Intelligence, Yonsei University.

Additional Skills

Programming Languages: Python, C/C++, SQL, Swift, JavaScript.

Deep Learning & Vision: PyTorch, TensorFlow.

Tools & Infrastructure: Git, Docker, Linux.

Languages: English and Kannada (Native); Korean (Business Proficiency, TOPIK 6); Hindi (Fluent).